STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 12: Kinetic Theory

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Negligible molecular volume and forces except during elastic collisions.

pV=nRT.

3k_BT/2.

Molecular momentum transfer to walls.

Three translational.

It increases.

3R/2.

sqrt(3RT/M).

Doubles.

About 300 K.

approx 6.21x10^-21 J.

2.

3nRT/2.

Negligible particle volume and negligible intermolecular forces except collisions.

approx 517 m/s.

approx 249 J.

One-third.

Three translational modes contribute 3k_BT/2 per molecule.

Lighter gas speed is twice as high.

U=(5/2)nRTapprox 12.5 kJ.